

Tianze Wu

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PROFILE

MSc Medical Microbiology graduate with hands-on laboratory experience in mammalian cell culture, bacterial culture, and in vitro experimental models. Demonstrated ability to perform routine laboratory procedures under SOPs, maintain accurate documentation, and support microbiology and cell-based research projects. Strong background in TEER assays, immunofluorescence staining, PCR workflows, and data analysis.

Eligible to work in the UK (2 year Graduate Visa valid until 2027)

EDUCATION

University of Manchester — MSc Medical Microbiology (Merit) Sep 2024— Sep 2025
Manchester, United Kingdom

China Pharmaceutical University — BSc in Pharmacy (Pharmacology & Biochemistry) Sep 2020— May 2024
Nanjing, China

- Selected coursework: Pharmacology; Pharmaceutics; Drug Analysis; Organic Chemistry; Physiology; Biochemistry; Bioinformatics; Molecular Biology; Cell Biology.
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RESEARCH EXPERIENCE

Assessment of the Cellular Actin Cytoskeleton During Disease 2024 — 2025
University of Manchester — Supervisor: Stavros Panagiotou Manchester, UK

- The Master's degree thesis
- Designed and executed a research project investigating the role of the cellular actin cytoskeleton in maintaining blood-brain barrier (BBB) integrity under pathological conditions
Established an in vitro BBB model using primary human brain microvascular endothelial cells (HBMECs) co-cultured with astrocytes on Transwell inserts to simulate physiological barrier properties. Evaluated barrier dysfunction by exposing the model to inflammatory, viral, and bacterial stimuli, with and without hypoxic stress. Measured transendothelial electrical resistance (TEER) over time and performed quantitative immunofluorescence analysis of F-actin and junctional proteins (e.g., ZO-1, occludin) to assess cytoskeletal remodeling and paracellular permeability. Demonstrated through two-way ANOVA statistical analysis that all stimuli induced significant BBB disruption via stress fiber formation and contractile retraction, with hypoxia consistently exacerbating damage and highlighting multi-hit injury mechanisms in neurological diseases. Identified actin regulatory pathways as potential therapeutic targets for preserving BBB function in complex disease contexts, including ischemic stroke superimposed on infection or inflammation
- Performing mammalian cell culture and maintenance under sterile conditions following laboratory SOPs.
- Performed immunofluorescence staining and fluorescence microscopy imaging.
- Analysed experimental data using GraphPad Prism and ImageJ/Fiji.
- Maintained detailed laboratory records, experimental logs, and sample documentation.
- Supported experimental optimisation, troubleshooting, and reproducibility checks.

Amplification and Extraction of L-asparaginase (AnsB) Protein Protein Expression & Microbial Techniques Project 2023 — 2024

Supervisor: Dr. Yi Pan

- Performed PCR amplification, plasmid construction, and bacterial transformation.
- Cultured E. coli under selective conditions using aseptic technique.
- Verified protein expression using SDS-PAGE.

- Conducted protein extraction and purification via ultrasonication and ion-exchange chromatography.
- Followed biosafety and waste disposal procedures in microbiology laboratory environments.

Prospects of Artificially Modified Benzene-ring Amino Acids for PKU Therapy (Microbial Genetics)

2023 — 2024

Supervisor: Dr. Xiaoda Song

- Conducted bacterial culture and growth assays in defined media (M9).
- Executed PCR workflows, including primer design and gene amplification.
- Performed competent cell preparation and transformation.
- Supported phenotype screening and resistance-based selection experiments.

PROFESSIONAL & RELEVANT EXPERIENCE

Lab technician

Nanjing Canchen Microbial Technology Co., Ltd.

2024 summer

Nanjing, China

- Supported laboratory-based feasibility studies related to inhaled antibiotic formulations.
- Assisted with experimental documentation and protocol review.
- Translated FDA and EMA regulatory materials to support experimental planning.

Teaching assistant (Microbiology Laboratory)

University of Manchester

2025 fall

Manchester, UK

- Demonstrated microbiology laboratory techniques to students.
- Reinforced laboratory safety, aseptic technique, and accurate documentation.
- Prepared written and practical teaching materials.

TECHNICAL SKILLS

Cell & Molecular Biology: • Mammalian cell culture • Bacterial culture and aseptic technique • Transwell assays • TEER measurement • Immunofluorescence staining • PCR, plasmid construction, bacterial transformation • SDS-PAGE, protein extraction and purification

Data & Analysis: • GraphPad Prism • ImageJ / Fiji • Excel • Statistical analysis and data visualisation

Laboratory Practice: • Operated within a GMP-aligned environment, adhering to SOP-controlled workflows, documentation requirements, and compliance standards • Sample preparation and handling

PUBLICATIONS & PRESENTATIONS

Wu, T. (in preparation). *Actin cytoskeleton remodelling and BBB integrity under disease-relevant stimuli in HBMECs*. [Manuscript planned post-thesis].

Design and Synthesis of NQO1 Responsive Fluorescence Probe and Its Application in Bio-imaging for Cancer Diagnosis

• **Journal:** Elsevier

• **Authors:** Zhenwei Yuan, Mingjun Xu, Tianze Wu, Xinyi Zhang, Yuanzhi Shen, Umeorah Ernest, Lijuan Gui, Fei Wang, Qing He, Haiyan Chen

• **URL:** <https://www.sciencedirect.com/science/article/abs/pii/S0039914019301523>

REFERENCES

Available upon request.